

Supplementary Document of “Hierarchical Sub-Array Beam Training for Flexible Intelligent Metasurface-Enabled Hybrid Near-Far-Field Multiuser Communications”

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Abstract

This supplementary document provides a rigorous analysis of the convergence, stationarity, optimality, and computational complexity of the algorithms developed for FIM-enabled hybrid near-far-field multiuser communications. For the optimization-based design, the complete alternating-optimization framework is examined by separately analyzing the closed-form desired-gain update, the proximal-distance-based analog-beamformer update, and the tanh-reparameterized surface-shaping update. Under standard smoothness conditions, summable PDA approximation errors, and Armijo backtracking, the original objective is shown to converge, while the continuous analog beamformers asymptotically approach the discrete phase set. The vanishing transformed-domain gradient and the corresponding boundary conditions further establish that any convergent solution is a block-wise steady point: the desired-gain block is globally optimal, the analog-beamformer block satisfies the PDA fixed-point condition, and the physical deformation satisfies the KKT conditions of the box-constrained subproblem. For the key-observation method, global optimality is proved for each simplified per-element quartic-polynomial problem and for the resulting separable surrogate, while its optimality scope and performance loss relative to the AO-based method are explicitly characterized. Finally, detailed complexity analysis shows that the rank-one structure of the PDA update avoids generic $N \times N$ matrix inversion, yielding linear per-iteration dependence on the numbers of FIM elements and users. KOM has an overall complexity of $\mathcal{O}(NK)$, whereas the subsequent WMMSE digital-beamforming stage has complexity $\mathcal{O}(NK^2 + I_W I_C K^3)$.

I. CONVERGENCE ANALYSIS OF THE ANALOG BEAMFORMER DESIGN ALGORITHM

The original analog beamformer design problem is given by

$$\begin{aligned} \min_{\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}} J(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}) &= \sum_{k=1}^K |\mathbf{a}_k(\Delta \mathbf{y})^H \mathbf{v}_k - g_k|^2 \\ \text{s.t. } &-c \leq \Delta y_n \leq c, |g_k| = N, \\ &[\mathbf{v}_k]_n = e^{j\tau_{k,n}}, \tau_{k,n} \in \mathcal{C}_b, k = 1, \dots, K, \end{aligned} \quad (1)$$

where $c = \hat{y}/2$, $\mathbf{a}_k(\Delta \mathbf{y}) = [e^{j\hat{\varphi}_k(\Delta y_1)}, e^{j\hat{\varphi}_k(\Delta y_2)}, \dots, e^{j\hat{\varphi}_k(\Delta y_N)}]^T$, $\hat{\varphi}_k(\Delta y_n) = \varphi_n(\Delta y_n)$ with respect to θ_k , and

$$\mathcal{V}_b = \{\mathbf{v} \in \mathbb{C}^N : [\mathbf{v}]_n = e^{j\tau_n}, \tau_n \in \mathcal{C}_b\} \quad (2)$$

denotes the finite feasible set associated with the discrete phase shifters (PS), where $\mathcal{C}_b \triangleq \{0, \frac{2\pi}{2^b}, \dots, \frac{(2^b-1)2\pi}{2^b}\}$.

The proposed alternating optimization (AO) algorithm sequentially updates the desired gains $\mathbf{g}$, the analog beamformers $\mathbf{V}$, and the surface shaping variables $\Delta \mathbf{y}$. The proximal-distance algorithm is employed only within the analog-beamformer block, while the tanh reparameterization is used only within the surface-deformation block. Let $J^{(t)} = J(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t)}, \mathbf{g}^{(t)})$ denote the original objective at the t -th outer AO iteration. Here, $\mathbf{V}^{(t)} \in \mathcal{V}_b^K$ denotes the discrete beamformer associated with the projection already used in the PDA procedure. The corresponding continuous auxiliary PDA variable is denoted by $\tilde{\mathbf{V}}^{(t)}$.

A. Assumptions

The convergence analysis is established under the following standard conditions.

Assumptions 1: The user-to-element distances are strictly positive. Consequently, $\mathbf{a}_k(\Delta \mathbf{y})$ and J are twice continuously differentiable with respect to $\Delta \mathbf{y}$ over the compact deformation set $[-c, c]^N$.

Assumptions 2: The desired-gain block is solved exactly, and the tanh-based deformation step employs an Armijo backtracking line search.

Assumptions 3: For the analog-beamformer block, let

$$d_{\mathcal{V}_b^K}(\tilde{\mathbf{V}}) = \min_{\mathbf{Q} \in \mathcal{V}_b^K} \|\tilde{\mathbf{V}} - \mathbf{Q}\|_F. \quad (3)$$

At outer iteration t , the PDA inner loop returns a continuous variable $\tilde{\mathbf{V}}^{(t+1)}$ satisfying

$$J\left(\Delta \mathbf{y}^{(t)}, \tilde{\mathbf{V}}^{(t+1)}, \mathbf{g}^{(t+1)}\right) + \mu_t \cdot d_{\mathcal{V}_b^K}^2\left(\tilde{\mathbf{V}}^{(t+1)}\right) \leq J\left(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t)}, \mathbf{g}^{(t+1)}\right) + \zeta_t, \quad (4)$$

where $\mu_t > 0$ is the final penalty factor used in the beamformer block and $\zeta_t \geq 0$ is its inner-solution error.

Condition (4) holds with $\zeta_t = 0$ if the corresponding penalized subproblem is minimized no worse than the previous feasible beamformer. For an inexact inner solution, ζ_t quantifies the residual solution error.

Assumptions 4: The penalty and inner-accuracy sequences satisfy

$$\mu_t \rightarrow \infty, \quad \sum_{t=0}^{\infty} \left(\frac{1}{\sqrt{\mu_t}} + \zeta_t \right) < \infty. \quad (5)$$

A sufficient choice is

$$\mu_t = \mu_0 \gamma^t, \quad \gamma > 1, \quad (6)$$

together with a summable inner error sequence, such as

$$\zeta_t \leq \zeta_0 \rho^t, \quad 0 < \rho < 1. \quad (7)$$

B. Desired-Gain Update

For fixed $\Delta \mathbf{y}^{(t)}$ and $\mathbf{V}^{(t)}$, define

$$\mathbf{s}_k^{(t)} = \mathbf{a}_k^H \left(\Delta \mathbf{y}^{(t)} \right) \mathbf{v}_k^{(t)}. \quad (8)$$

The desired-gain subproblem is

$$\min_{\mathbf{g}} \sum_{k=1}^K \left| s_k^{(t)} - g_k \right|^2 \quad \text{s.t. } |g_k| = N, \quad (9)$$

The subproblem is separable over k , and its globally optimal solution is

$$g_k^{(t+1)} = N e^{j \angle(s_k^{(t)})}. \quad (10)$$

If $s_k^{(t)} = 0$, any g_k satisfying $|g_k| = N$ is optimal. Therefore, we have

$$J\left(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t)}, \mathbf{g}^{(t+1)}\right) \leq J\left(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t)}, \mathbf{g}^{(t)}\right) = J^{(t)}. \quad (11)$$

Hence, the desired-gain update does not increase the original objective.

C. PDA-Based Analog-Beamformer Update

For fixed $\Delta \mathbf{y}^{(t)}$ and $\mathbf{g}^{(t+1)}$, define

$$F_t(\mathbf{V}) = J\left(\Delta \mathbf{y}^{(t)}, \mathbf{V}, \mathbf{g}^{(t+1)}\right). \quad (12)$$

The original beamformer subproblem is

$$\min_{\mathbf{V} \in \mathcal{V}_b^K} F_t(\mathbf{V}). \quad (13)$$

The PDA instead considers the penalized continuous problem

$$\min_{\tilde{\mathbf{V}}} F_t(\tilde{\mathbf{V}}) + \mu_t \cdot d_{\mathcal{V}_b^K}^2(\tilde{\mathbf{V}}). \quad (14)$$

Let

$$\mathbf{V}^{(t+1)} \in \Pi_{\mathcal{V}_b^K} \left(\tilde{\mathbf{V}}^{(t+1)} \right) \quad (15)$$

be a nearest discrete projection.

1) Asymptotic discrete feasibility

Since the original feasible set is compact and finite, and $\Delta \mathbf{y}$ and $\mathbf{g}$ are bounded, there exists a finite constant M_J such that

$$0 \leq J(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}) \leq M_J \quad (16)$$

for all feasible $(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g})$.

It follows from (4) that

$$\mu_t \cdot d_{\mathcal{V}_b^K}^2(\tilde{\mathbf{V}}^{(t+1)}) \leq F_t(\mathbf{V}^{(t)}) + \zeta_t \leq M_J + \zeta_t. \quad (17)$$

Therefore, we have

$$d_{\mathcal{V}_b^K}(\tilde{\mathbf{V}}^{(t+1)}) \leq \sqrt{\frac{M_J + \zeta_t}{\mu_t}}. \quad (18)$$

Under Assumption 4, we can obtain

$$\left\| \tilde{\mathbf{V}}^{(t+1)} - \mathbf{V}^{(t+1)} \right\|_{\text{F}} \rightarrow 0. \quad (19)$$

Thus, the PDA continuous iterate becomes asymptotically identical to a feasible discrete beamformer.

2) Quasi-descent of the original beamforming objective

Since $F_t(\mathbf{V})$ is continuously differentiable and all relevant beamformer iterates remain in a bounded neighborhood of the finite set $\mathcal{V}_b^K$, there exists a uniform constant $L_V > 0$ such that

$$|F_t(\mathbf{V}_1) - F_t(\mathbf{V}_2)| \leq L_V \|\mathbf{V}_1 - \mathbf{V}_2\|_{\text{F}}. \quad (20)$$

Using (15), (18), and (20), we obtain

$$F_t(\mathbf{V}^{(t+1)}) \leq F_t(\tilde{\mathbf{V}}^{(t+1)}) + L_V \left\| \tilde{\mathbf{V}}^{(t+1)} - \mathbf{V}^{(t+1)} \right\|_{\text{F}}. \quad (21)$$

Moreover, from (4),

$$F_t(\tilde{\mathbf{V}}^{(t+1)}) \leq F_t(\mathbf{V}^{(t)}) + \zeta_t. \quad (22)$$

Combining (18), (21), and (22) yields

$$F_t(\mathbf{V}^{(t+1)}) \leq F_t(\mathbf{V}^{(t)}) + \epsilon_t, \quad (23)$$

where $\epsilon_t = \zeta_t + L_V \sqrt{(M_J + \zeta_t)/\mu_t}$.

Because of Assumption 4, we have

$$\sum_{t=0}^{\infty} \epsilon_t < \infty. \quad (24)$$

Hence, the PDA block satisfies a quasi-descent condition for the original objective, with an error that vanishes and is summable. In the ideal case where the PDA continuation is run to the limit $\mu_t \rightarrow \infty$, $\zeta_t \rightarrow 0$ within each beamformer update, one has $\epsilon_t = 0$ in the limiting sense, and (23) reduces to the exact descent relation

$$F_t(\mathbf{V}^{(t+1)}) \leq F_t(\mathbf{V}^{(t)}). \quad (25)$$

D. tanh-Based Surface Shaping Update

For fixed $\mathbf{V}^{(t+1)}$ and $\mathbf{g}^{(t+1)}$, define the surface shaping subproblem

$$\min_{\Delta \mathbf{y}} J(\Delta \mathbf{y}, \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)}) \quad \text{s.t.} \quad -c \leq \Delta y_n \leq c. \quad (26)$$

Introduce $\Delta y_n = h_n(z_n) = c \cdot \tanh(\alpha z_n)$, $z_n \in \mathbb{R}$ and define $\mathbf{h}(\mathbf{z}) = c \cdot \tanh(\alpha \mathbf{z})$, $\varphi_t(\mathbf{z}) = J(\mathbf{h}(\mathbf{z}), \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)})$. The Jacobian of the tanh mapping is

$$\mathbf{D}(\mathbf{z}) = \text{diag} \left\{ c \cdot \alpha \cdot \text{sech}^2(\alpha z_n) \right\}_{n=1}^N. \quad (27)$$

By the chain rule, we have

$$\nabla_{\mathbf{z}} \varphi_t(\mathbf{z}) = \mathbf{D}(\mathbf{z}) \cdot \nabla_{\Delta \mathbf{y}} J(\mathbf{h}(\mathbf{z}), \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)}). \quad (28)$$

The update is $\mathbf{z}^{(t+1)} = \mathbf{z}^{(t)} - \eta_t \nabla_{\mathbf{z}} \varphi(\mathbf{z}^{(t)})$. The step size is selected by Armijo backtracking:

$$\eta_t = \bar{\eta} \beta^{m_t}, \quad 0 < \beta < 1, \quad (29)$$

where m_t is the smallest non-negative integer satisfying

$$\varphi_t \left(\mathbf{z}^{(t)} - \eta_t \nabla \varphi_t \left(\mathbf{z}^{(t)} \right) \right) \leq \varphi_t(\mathbf{z}^{(t)}) - \sigma \cdot \eta_t \cdot \left\| \nabla \varphi_t(\mathbf{z}^{(t)}) \right\|_2^2, \quad (30)$$

with $0 < \sigma < 1$.

Thus, we can obtain

$$J \left(\Delta \mathbf{y}^{(t+1)}, \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)} \right) \leq J \left(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)} \right) - \sigma \cdot \eta_t \cdot \left\| \nabla_{\mathbf{z}} \varphi_t \left(\mathbf{z}^{(t)} \right) \right\|_2^2. \quad (31)$$

E. Convergence of the Original Objective

Proposition 1: Under Assumptions 1–4, the sequence of original objective values $\{J^{(t)}\}$ generated by the proposed AO algorithm converges to a finite value.

Proof:

Combining the exact desired-gain descent in (11), the PDA quasi-descent in (23), and the tanh-gradient sufficient decrease in (31), we obtain

$$J^{(t+1)} \leq J^{(t)} + \epsilon_t - \sigma \cdot \eta_t \left\| \nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)}) \right\|_2^2. \quad (32)$$

Define the tail error

$$E_t = \sum_{i=t}^{\infty} \epsilon_i. \quad (33)$$

By (24), $E_t < \infty$ and $E_t \rightarrow 0$. Define the corrected objective sequence $\hat{J}^{(t)} = J^{(t)} + E_t$. Since $E_t = \epsilon_t + E_{t+1}$, inequality (32) yields

$$\hat{J}^{(t+1)} = J^{(t+1)} + E_{t+1} \leq J^{(t)} + \epsilon_t + E_{t+1} - \sigma \cdot \eta_t \cdot \left\| \nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)}) \right\|_2^2 = \hat{J}^{(t)} - \sigma \cdot \eta_t \cdot \left\| \nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)}) \right\|_2^2. \quad (34)$$

Thus, $\{\hat{J}^{(t)}\}$ is monotonically non-increasing. Moreover, $\hat{J}^{(t)} \geq 0$, and hence it converges to a finite value. Since $E_t \rightarrow 0$, it follows that the original objective also converges:

$$J^{(t)} \rightarrow J^* < \infty. \quad (35)$$

The completes the proof. ■

II. STEADY-POINT ANALYSIS OF THE ANALOG BEAMFORMER DESIGN ALGORITHM

Corollary 1: Vanishing tanh-Domain Gradient

Summing (34) over t gives

$$\sum_{t=0}^{\infty} \sigma \cdot \eta_t \cdot \left\| \nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)}) \right\|_2^2 < \infty. \quad (36)$$

Since the physical surface shaping, the discrete beamformers, and the desired gains lie in compact sets, and the first and second derivatives of the tanh mapping are bounded, there exists a uniform Lipschitz constant L_z for $\nabla \varphi_t$.

The Armijo procedure therefore accepts a step size bounded away from zero:

$$\eta_t \geq \eta_{\min} > 0. \quad (37)$$

Consequently, we can obtain

$$\lim_{t \rightarrow \infty} \left\| \nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)}) \right\|_2 = 0. \quad (38)$$

Furthermore, we have

$$\left\| \Delta \mathbf{y}^{(t+1)} - \Delta \mathbf{y}^{(t)} \right\|_2 \leq c \cdot \alpha \cdot \left\| \mathbf{z}^{(t+1)} - \mathbf{z}^{(t)} \right\|_2 = c \cdot \alpha \cdot \left\| \nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)}) \right\|_2, \quad (39)$$

which gives

$$\left\| \Delta \mathbf{y}^{(t+1)} - \Delta \mathbf{y}^{(t)} \right\|_2 \rightarrow 0. \quad (40)$$

A. Existence of Accumulation Points

The physical deformation satisfies

$$\Delta \mathbf{y}^{(t)} \in [-c, c]^N. \quad (41)$$

The desired-gain variables satisfy $|g_k^{(t)}| = N$, and the projected analog beamformers satisfy $\mathbf{V}^{(t)} \in \mathcal{V}_b^K$. Therefore, the physical feasible set is compact. Hence, the sequence $\{\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t)}, \mathbf{g}^{(t)}\}$ has at least one accumulation point.

Objective-value convergence does not necessarily imply uniqueness of the accumulation point. The following result therefore characterizes any convergent sequence of physical AO variables.

B. Steady-Point Characterization

Proposition 2: Suppose that the physical AO sequence converges $(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t)}, \mathbf{g}^{(t)}) \rightarrow (\Delta \mathbf{y}^*, \mathbf{V}^*, \mathbf{g}^*)$. Then the limit satisfies the following block-wise steady-state conditions.

1) Desired-gain block

Because the desired-gain subproblem is solved globally at every iteration,

$$\mathbf{g}^* = \arg \min_{\mathbf{g}} J(\Delta \mathbf{y}^*, \mathbf{V}^*, \mathbf{g}) \quad \text{s.t. } |g_k| = N. \quad (42)$$

More specifically, when $\mathbf{a}_k^H(\Delta \mathbf{y}^*) \mathbf{v}_k^* \neq 0$, we have $g_k^* = N \cdot e^{j\angle(\mathbf{a}_k^H(\Delta \mathbf{y}^*) \mathbf{v}_k^*)}$.

2) Analog-beamformer block

Equation (19) implies

$$\|\tilde{\mathbf{V}}^{(t)} - \mathbf{V}^{(t)}\|_F \rightarrow 0. \quad (43)$$

Hence, we have

$$\tilde{\mathbf{V}}^{(t)} \rightarrow \mathbf{V}^*, \quad \mathbf{V}^* \in \mathcal{V}_b^K. \quad (44)$$

Therefore, the limiting beamformer is feasible for the original discrete phase constraint.

Let $\mathcal{T}_{\mu_t}(\tilde{\mathbf{V}}; \Delta \mathbf{y}, \mathbf{g})$ denote the closed-form PDA update mapping. If the PDA inner residual satisfies

$$\|\tilde{\mathbf{V}}^{(t+1)} - \mathcal{T}_{\mu_t}(\tilde{\mathbf{V}}^{t+1}; \Delta \mathbf{y}^{(t)}, \mathbf{g}^{(t+1)})\|_F \rightarrow 0, \quad (45)$$

then $\mathbf{V}^*$ is an asymptotically feasible fixed point of the PDA beamformer update.

3) Surface shaping block

Define $r_{y,n}^* = \frac{\partial J}{\partial \Delta y_n}(\Delta \mathbf{y}^*, \mathbf{V}^*, \mathbf{g}^*)$.

Interior components

If $-c < \Delta y_n^* < c$, then $z_n^* = \frac{1}{\alpha} \operatorname{artanh}\left(\frac{\Delta y_n^*}{c}\right)$ is finite, and $c \cdot \alpha \cdot \operatorname{sech}^2(\alpha z_n^*) > 0$.

Using (28) and (38), we obtain

$$r_{y,n}^* = 0, \quad -c < \Delta y_n^* < c. \quad (46)$$

Upper-bound components

If $\Delta y_n^* = c$, then $z_n^{(t)} \rightarrow +\infty$. Suppose, by contradiction, that $r_{y,n}^* > 0$.

By continuity, there exist $\delta > 0$ and T such that

$$\frac{\partial J}{\partial \Delta y_n}(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)}) \geq \delta, \quad t \geq T. \quad (47)$$

The update of the n -th auxiliary variable then satisfies

$$z_n^{(t+1)} - z_n^{(t)} = -\eta_t \cdot c \cdot \alpha \cdot \operatorname{sech}^2(\alpha z_n^{(t)}) \times \frac{\partial J}{\partial \Delta y_n}(\Delta \mathbf{y}^{(t)}, \mathbf{V}^{(t+1)}, \mathbf{g}^{(t+1)}) < 0. \quad (48)$$

Thus, $z_n^{(t)}$ is eventually decreasing and cannot approach $+\infty$, yielding a contradiction. Therefore,

$$r_{y,n}^* \leq 0, \quad \Delta y_n^* = c. \quad (49)$$

Lower-bound components

Similarly, if $\Delta y_n^* = -c$, then $z_n^{(t)} \rightarrow -\infty$. Assuming $r_{y,n}^* < 0$ would imply that $z_n^{(t)}$ is eventually increasing and therefore cannot approach $-\infty$. Hence,

$$r_{y,n}^* \geq 0, \quad \Delta y_n^* = -c. \quad (50)$$

Combining (46), (49), and (50), the limiting deformation satisfies

$$\begin{cases} \frac{\partial J^*}{\partial \Delta y_n} = 0, & -c < \Delta y_n^* < c, \\ \frac{\partial J^*}{\partial \Delta y_n} \leq 0, & \Delta y_n^* = c, \\ \frac{\partial J^*}{\partial \Delta y_n} \geq 0, & \Delta y_n^* = -c. \end{cases} \quad (51)$$

Equivalently, $\mathbf{0} \in \nabla_{\Delta \mathbf{y}} J(\Delta \mathbf{y}^*, \mathbf{V}^*, \mathbf{g}^*) + N_{[-c,c]^N}(\Delta \mathbf{y}^*)$, where $N_{[-c,c]^N}$ denotes the normal cone of the deformation box. Thus, the limiting physical deformation satisfies the KKT conditions of the original box-constrained deformation subproblem.

C. Practical Stopping Criteria

The complete AO algorithm may be terminated when

$$\frac{|J^{(t+1)} - J^{(t)}|}{\max(1, J^{(t)})} \leq \epsilon_J, \quad (52)$$

$$\frac{\|\tilde{\mathbf{V}}^{(t+1)} - \Pi_{\mathcal{V}_b^K}(\tilde{\mathbf{V}}^{(t+1)})\|_{\text{F}}}{\sqrt{KN}} \leq \epsilon_V, \quad (53)$$

and

$$\frac{\nabla_{\mathbf{z}} \varphi_t(\mathbf{z}^{(t)})}{\sqrt{N}} \leq \epsilon_z, \quad (54)$$

or when the prescribed maximum number of AO iterations is reached.

For every fixed PDA penalty factor, the exact desired-gain update, the majorization-based PDA beamformer update, and the Armijo-controlled tanh deformation update monotonically decrease the same merit function. Since this function is nonnegative, its value converges. The desired-gain optimality gap, the beamformer-update difference, and the transformed deformation-gradient norm all vanish asymptotically. Consequently, every accumulation point is globally optimal with respect to the desired-gain block, satisfies the PDA fixed-point condition for the analog-beamformer block, and satisfies the KKT condition of the box-constrained deformation block. Under penalty continuation, the beamformer distance to the finite discrete phase set vanishes, and every accumulation point is therefore a block-wise steady point of the original mixed discrete-continuous problem. Owing to the nonconvexity and the finite phase constraints, global optimality is not guaranteed.

D. Simulation Results

Fig. 1 illustrates the convergence behavior of the original objective value with respect to the outer AO iteration index. The objective decreases sharply during the first few iterations and subsequently remains nearly unchanged. This result demonstrates that the sequential updates of the desired gains, analog beamformers, and FIM deformation variables rapidly reduce the multi-beam fitting error and drive the AO objective toward a stable value.

Fig. 2 further examines the convergence and stationarity of the tanh-based surface-shaping update. As shown in Fig. 2(a), the transformed-domain gradient norm decreases rapidly during the initial iterations and approaches zero as the iteration proceeds, numerically confirming the first-order stationarity condition. Meanwhile, Fig. 2(b) shows that the norm of the difference between two consecutive physical deformation iterates also gradually vanishes. This behavior verifies the asymptotic regularity of the deformation sequence and indicates that the surface shape no longer changes appreciably after convergence. Together, the two curves provide numerical evidence that the proposed tanh-based update converges to a first-order steady solution rather than merely producing a stable objective value.

Fig. 3 demonstrates the convergence of the continuous PDA beamformer toward the discrete phase set. The distance between the continuous beamformer and its nearest quantized projection decreases rapidly by several orders of magnitude and eventually approaches zero. The enlarged view further shows that the small residual distance continues to decrease during the later iterations. This result numerically verifies that increasing the PDA penalty factor progressively suppresses the discrete-constraint violation and makes the continuous PDA solution asymptotically consistent with a feasible quantized beamformer. It should be noted that this figure verifies asymptotic discrete feasibility.

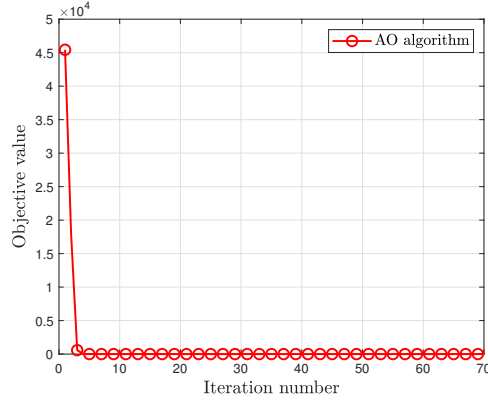


Fig. 1. Convergence of the original objective value for the proposed AO algorithm.

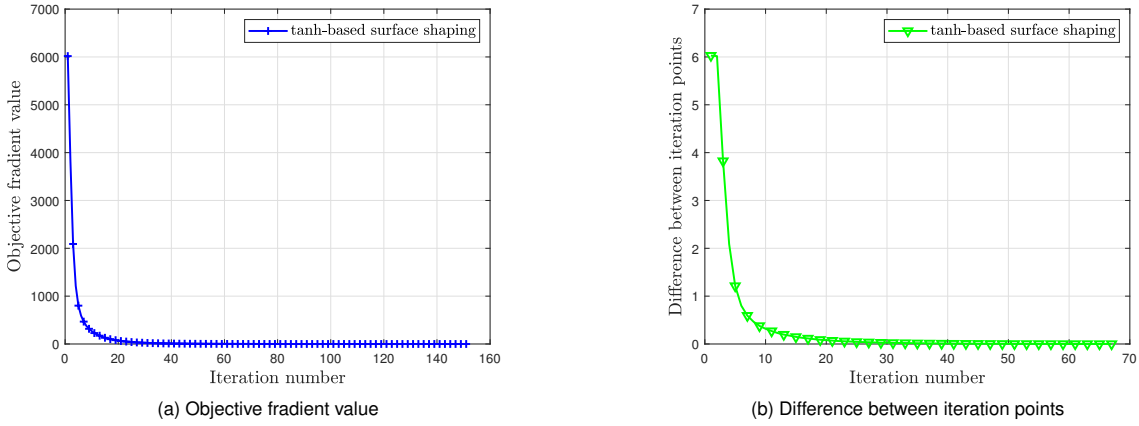


Fig. 2. Convergence behavior of the tanh-based surface-shaping update: (a) transformed-domain gradient norm; (b) norm of the difference between consecutive physical deformation iterates.

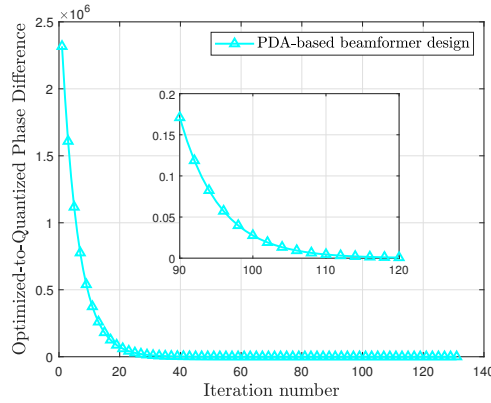


Fig. 3. Convergence of the continuous PDA beamformer toward the discrete phase set. The inset shows the residual projection distance during the later iterations.

III. OPTIMALITY AND PERFORMANCE CHARACTERIZATION OF KOM

We propose to shape the FIM surface such that the phase quantization errors across all user steering vectors are minimized, a strategy we refer to as the key observation method (KOM). Based on this, the surface shape of the FIM can be obtained by simultaneously solving N problems as follows

$$\min_{\Delta y_n \in [-\frac{\hat{y}}{2}, \frac{\hat{y}}{2}]} \sum_{k=1}^K (A_{n,k} \Delta y_n^2 + B_{n,k} \Delta y_n + C_{n,k})^2, \quad (55)$$

where

$$A_{n,k} = \frac{-k_c \sin^2 \theta_k}{2r_k}, \quad B_{n,k} = \frac{k_c \cos \theta_k (\delta_n d \sin \theta_k + r_k)}{r_k}.$$

When $\tilde{\varphi}_k(0) \leq Q_b(\tilde{\varphi}_k(0))$, $C_{n,k} = -|\tilde{\varphi}_k(0) - Q_b(\tilde{\varphi}_k(0))|$; otherwise, $C_{n,k} = |\tilde{\varphi}_k(0) - Q_b(\tilde{\varphi}_k(0))|$, where $\tilde{\varphi}_k(0) = \text{mod}(\varphi_k(0), 2\pi)$.

Then, the problem (55) can be reformulated as follows

$$\min_{\Delta y_n \in [-\frac{y}{2}, \frac{y}{2}]} \mathcal{G}_n(\Delta y_n) = \alpha_{4,n} \Delta y_n^4 + \alpha_{3,n} \Delta y_n^3 + \alpha_{2,n} \Delta y_n^2 + \alpha_{1,n} \Delta y_n + \alpha_{0,n}, \quad (56)$$

where $\alpha_{0,n} = \sum_{k=1}^K C_{n,k}^2$, $\alpha_{1,n} = 2 \sum_{k=1}^K B_{n,k} C_{n,k}$, $\alpha_{2,n} = \sum_{k=1}^K (B_{n,k}^2 + 2A_{n,k} C_{n,k})$, $\alpha_{3,n} = 2 \sum_{k=1}^K A_{n,k} B_{n,k}$, and $\alpha_{4,n} = \sum_{k=1}^K A_{n,k}^2$.

The first derivative of $\mathcal{G}_n(\Delta y_n)$ is

$$\mathcal{G}'_n(\Delta y_n) = 4\alpha_{4,n} \Delta y_n^3 + 3\alpha_{3,n} \Delta y_n^2 + 2\alpha_{2,n} \Delta y_n + \alpha_{1,n}. \quad (57)$$

Let

$$\mathcal{R}_n = \left\{ \Delta y_n \in \mathbb{R} : \mathcal{G}'_n(\Delta y_n) = 0 \right\} \quad (58)$$

denote the set of real stationary points of the quartic polynomial, and define the feasible candidate set as

$$\mathcal{S}_n = \{-c, c\} \cup (\mathcal{R}_n \cap [-c, c]). \quad (59)$$

Since a cubic equation has at most three real roots, $\mathcal{S}_n$ contains at most five candidate points. The KOM solution is then obtained as

$$\Delta y_n^{\text{KOM}} = \arg \min_{\Delta y_n \in \mathcal{S}_n} \mathcal{G}_n(\Delta y_n). \quad (60)$$

When several candidates yield the same minimum value, any of them can be selected.

A. Global Optimality of KOM

1) Global Optimality of Per-Element Problem

Proposition 3: The solution Δy_n^{KOM} obtained from (60) is a globally optimal solution of the simplified per-element problem in (55).

Proof:

The objective $\mathcal{G}_n(\Delta y_n)$ is a polynomial and is therefore continuous over the compact interval $[-c, c]$. According to the Weierstrass theorem, the global minimum of $\mathcal{G}_n(\Delta y_n)$ over this interval exists. Let Δy_n^* denote a global minimizer. Two cases need to be considered. If $-c < \Delta y_n^* < c$, then Δy_n^* is an interior minimizer. According to Fermat's first-order necessary condition, $\mathcal{G}'_n(\Delta y_n^*) = 0$. Therefore, $\Delta y_n^* \in \mathcal{R}_n \cap [-c, c]$. Otherwise, the global minimizer lies on the boundary of the feasible interval, i.e., $\Delta y_n^* = -c$ or $\Delta y_n^* = c$.

Consequently, every global minimizer belongs to the candidate set $\mathcal{S}_n$. Evaluating $\mathcal{G}_n(\Delta y_n)$ at all feasible stationary points and the two interval boundaries therefore identifies a global minimizer of (55). Hence, the solution in (59) is globally optimal for the simplified per-element polynomial problem.

The proof is complete. ■

Remark 1: Degenerate Cases

The above result also applies when the quartic polynomial degenerates into a cubic, quadratic, linear, or constant function. In these cases, the derivative equation in (56) has a degree lower than three. The global minimizer is still obtained by comparing all feasible real stationary points with the two interval boundaries.

2) Global Optimality for the Separable KOM Surrogate

The complete KOM surrogate can be expressed as

$$\mathcal{G}(\Delta \mathbf{y}) = \sum_{n=1}^N \mathcal{G}_n(\Delta y_n), \quad (61)$$

with the Cartesian-product feasible set

$$\mathcal{Y} = [-c, c]^N. \quad (62)$$

Because both the objective and the feasible set are separable across the radiating elements,

$$\min_{\Delta \mathbf{y} \in \mathcal{Y}} = \min_{\Delta y_1, \dots, \Delta y_N} \sum_{n=1}^N \mathcal{G}_n(\Delta y_n) = \sum_{n=1}^N \min_{\Delta y_n \in [-c, c]} \mathcal{G}_n(\Delta y_n). \quad (63)$$

Therefore, defining

$$\Delta \mathbf{y}^{\text{KOM}} = [\Delta y_1^{\text{KOM}}, \dots, \Delta y_N^{\text{KOM}}]^T, \quad (64)$$

we obtain

$$\Delta \mathbf{y}^{\text{KOM}} = \arg \min_{\Delta \mathbf{y} \in \mathcal{Y}} \mathcal{G}(\Delta \mathbf{y}). \quad (65)$$

Thus, KOM is globally optimal not only for each scalar polynomial problem but also for the resulting separable surrogate problem.

B. Scope of the Optimality Result

The global-optimality result in Propositions 3 and (65) applies only to the simplified KOM surrogate. The original surface-shaping objective has the form

$$J(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}) = \sum_{k=1}^K \left| \sum_{n=1}^N v_{n,k} e^{-j\varphi_k(\Delta y_n)} - g_k \right|^2. \quad (66)$$

Expanding the squared magnitude produces cross-element terms of the form

$$v_{n,k} \cdot v_{m,k}^* \cdot e^{-j[\varphi_k(\Delta y_n) - \varphi_k(\Delta y_m)]}, \quad n \neq m, \quad (67)$$

which couple the deformation variables of different radiating elements. The original objective is therefore generally non-separable and non-convex.

In contrast, KOM replaces the coupled objective by independent scalar polynomial objectives. Hence, *KOM is globally optimal for the simplified separable surrogate, but it is not guaranteed to be globally optimal for the original coupled problem.*

The optimization-based AO method, on the other hand, directly updates the original coupled objective but generally converges only to a block-wise stationary point. Therefore, neither method should be described as providing the global optimum of the original joint problem.

C. Performance Loss Relative to the AO-Based Method

1) Approximation-Induced Performance Bound

A conditional bound can be established to relate the KOM surrogate to the original deformation objective. For fixed $\mathbf{V}$ and $\mathbf{g}$, denote the original surface shaping objective by

$$\Phi(\Delta \mathbf{y}) = J(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}). \quad (68)$$

Suppose that the KOM surrogate satisfies a uniform approximation-error bound

$$|\Phi(\Delta \mathbf{y}) - \mathcal{G}(\Delta \mathbf{y})| \leq \epsilon_{\text{app}}, \quad \forall \Delta \mathbf{y} \in \mathcal{Y}. \quad (69)$$

Let

$$\Delta \mathbf{y}_{\Phi}^* = \arg \min_{\Delta \mathbf{y} \in \mathcal{Y}} \Phi(\Delta \mathbf{y}) \quad (70)$$

be a global minimizer of the original deformation subproblem, and let $\Delta \mathbf{y}^{\text{KOM}}$ be the global minimizer of the surrogate. Then,

$$\Phi(\Delta \mathbf{y}^{\text{KOM}}) \leq \mathcal{G}(\Delta \mathbf{y}^{\text{KOM}}) + \epsilon_{\text{app}} \leq \mathcal{G}(\Delta \mathbf{y}_{\Phi}^*) + \epsilon_{\text{app}} \leq \Phi(\Delta \mathbf{y}_{\Phi}^*) + 2\epsilon_{\text{app}}. \quad (71)$$

This result shows that the loss caused by optimizing the KOM surrogate is controlled by its uniform approximation error. If the approximation error can be bounded element-wise as

$$|\Phi_n(\Delta y_n) - \mathcal{G}_n(\Delta y_n)| \leq \epsilon_n, \quad \Delta y_n \in [-c, c], \quad (72)$$

then $\epsilon_{\text{app}} \leq \sum_{n=1}^N \epsilon_n$, and

$$\Phi(\Delta \mathbf{y}^{\text{KOM}}) - \Phi(\Delta \mathbf{y}_{\Phi}^*) \leq 2 \sum_{n=1}^N \epsilon_n. \quad (73)$$

A numerical performance comparison is still necessary unless explicit approximation-remainder bounds ϵ_n are available.

2) Performance Loss Analysis

Let

$$J_{\text{AO}} = J(\Delta \mathbf{y}_{\text{AO}}, \mathbf{V}_{\text{AO}}, \mathbf{g}_{\text{AO}}) \quad (74)$$

and

$$J_{\text{KOM}} = J(\Delta \mathbf{y}_{\text{KOM}}, \mathbf{V}_{\text{KOM}}, \mathbf{g}_{\text{KOM}}) \quad (75)$$

denote the original objective values achieved by the AO method and KOM method, respectively.

Since the objective is minimized, the signed relative objective gap is defined as

$$\text{Gap}_J = \frac{J_{\text{KOM}} - J_{\text{AO}}}{\max\{J_{\text{AO}}, \epsilon_0\}} \times 100\%, \quad (76)$$

where ϵ_0 is a small constant preventing numerical instability when J_{AO} is close to zero.

A positive Gap_J indicates that KOM yields a larger fitting error, whereas a negative value means that KOM attains a smaller objective than the particular local solution returned by the AO method. The latter does not contradict the analysis because the AO method is not globally optimal.

If achievable rate is the primary communication metric, define

$$\text{Loss}_R = \frac{R_{\text{AO}} - R_{\text{KOM}}}{R_{\text{AO}}} \times 100\%. \quad (77)$$

The absolute rate loss is

$$\Delta R = R_{\text{AO}} - R_{\text{KOM}}. \quad (78)$$

For the average beamforming gain, let

$$\bar{G}_{\mathcal{M}} = \frac{1}{K} \sum_{k=1}^K G_{k,\mathcal{M}}, \quad \mathcal{M} \in \{\text{AO}, \text{KOM}\}. \quad (79)$$

The gain loss in decibels can be defined as

$$\Delta G_{\text{dB}} = 10 \log_{10} \left(\frac{\bar{G}_{\text{AO}}}{\bar{G}_{\text{KOM}}} \right) \quad (80)$$

The computational advantage is quantified through

$$\text{Speedup} = \frac{T_{\text{AO}}}{T_{\text{KOM}}}, \quad (81)$$

where T_{AO} and T_{KOM} denote the average execution times under the same hardware and software environment.

IV. ALGORITHM COMPLEXITY ANALYSIS

A. Algorithm complexity analysis of the AO algorithm

We analyze the computational complexity of the optimization-based surface-shaping method, KOM, and the subsequent WMMSE digital beamforming algorithm. Let N , K , and $R = N_{\text{RF}}$ denote the numbers of FIM radiating elements, users, and RF chains, respectively. Let I_{AO} , I_{P} , and I_{G} denote the numbers of outer AO iterations, total PDA iterations within each AO iteration, and tanh-based gradient iterations within each AO iteration, respectively. Moreover, let I_{LS} denote the average number of objective evaluations required by the backtracking line search. For the optimization-based surface-shaping method, updating the desired gains requires calculating K length- N inner products, resulting in a complexity of

$$\mathcal{O}(NK). \quad (82)$$

For the analog-beamformer block, the projection onto the uniformly quantized phase set and the associated vector operations both require $\mathcal{O}(N)$ operations for each beamformer. Although the closed-form PDA update contains an $N \times N$ matrix expression, its rank-one structure allows it to be evaluated without explicitly forming or inverting a dense matrix. Specifically, the update can be implemented using one inner product and several vector operations, resulting in a complexity of $\mathcal{O}(N)$ per beamformer and

$$\mathcal{O}(I_{\text{P}}NK). \quad (83)$$

for the complete PDA block. Here, I_P includes all PDA inner iterations and penalty-factor updates. For a uniformly spaced b -bit phase set, the projection is implemented through nearest-neighbor quantization and has linear complexity in N .

For the surface shaping block, evaluating the K residuals requires summing the contributions of all N radiating elements, and hence has complexity $\mathcal{O}(NK)$. After the residuals are obtained, calculating the complete deformation gradient also requires $\mathcal{O}(NK)$ operations. Each backtracking trial reevaluates the objective with complexity $\mathcal{O}(NK)$. Therefore, the complexity of the deformation block is

$$\mathcal{O}(I_G(1 + I_{LS})NK). \quad (84)$$

Combining the three blocks, the overall complexity of the optimization-based surface-shaping method is

$$\mathcal{O}_{AO} = \mathcal{O}(I_{AO}NK[1 + I_P + I_G(1 + I_{LS})]). \quad (85)$$

B. Algorithm complexity analysis of the KOM method

For KOM, constructing the coefficients of the quartic polynomial associated with each radiating element requires summations over the K beams and therefore has complexity $\mathcal{O}(K)$. The derivative of each quartic polynomial is a cubic equation with at most three real roots. Solving this cubic equation and comparing its feasible real roots with the two interval boundaries involve a constant number of operations independent of N and K . Consequently, the total complexity of KOM is

$$\mathcal{O}_{KOM} = \mathcal{O}(NK). \quad (86)$$

Unlike the optimization-based method, KOM does not contain AO, PDA, gradient, or line-search iterations. It therefore achieves linear complexity with respect to both the number of FIM elements and the number of users.

C. Algorithm complexity analysis of the digital beamformer design

We next analyze the complexity of the WMMSE digital beamforming algorithm. Computing the effective user channels $\tilde{\mathbf{h}}_k^H = \mathbf{h}_k^H \mathbf{V}$ for all users requires $\mathcal{O}(NN_{\text{RF}}K)$ operations, while calculating $\mathbf{V}^H \mathbf{V}$ requires $\mathcal{O}(NN_{\text{RF}}^2)$ operations. Both quantities can be calculated once before the WMMSE iterations.

During each WMMSE iteration, evaluating all desired-signal and interference terms $\tilde{\mathbf{h}}_k^H \mathbf{w}_i$ requires $\mathcal{O}(N_{\text{RF}}K^2)$ operations. Constructing the weighted channel covariance matrix

$$\mathbf{E} = \sum_{k=1}^K \rho_k |u_k|^2 \tilde{\mathbf{h}}_k \tilde{\mathbf{h}}_k^H \quad (87)$$

requires

$$\mathcal{O}(KN_{\text{RF}}^2) \quad (88)$$

operations. For a given dual variable ζ , updating the digital beamformers requires factorizing an $N_{\text{RF}} \times N_{\text{RF}}$ matrix and solving K linear systems, whose complexity is

$$\mathcal{O}(N_{\text{RF}}^2 K + N_{\text{RF}}^3). \quad (89)$$

Let I_ζ denote the number of one-dimensional search iterations used to determine ζ , and let I_W denote the number of WMMSE iterations. The overall WMMSE complexity is therefore

$$\mathcal{O}_{\text{WMMSE}} = \mathcal{O}(NN_{\text{RF}}K + NN_{\text{RF}}^2 + I_W[N_{\text{RF}}K^2 + KN_{\text{RF}}^2 + I_\zeta(N_{\text{RF}}^3 + N_{\text{RF}}^2 K)]). \quad (90)$$

For the considered hybrid architecture, the number of RF chains is equal to the number of served users, i.e., $N_{\text{RF}} = K$. In this case, (90) reduces to

$$\mathcal{O}_{\text{WMMSE}} = \mathcal{O}(NK^2 + I_W I_\zeta K^3). \quad (91)$$

When I_ζ is regarded as a fixed small constant, the complexity can be written more compactly as

$$\mathcal{O}_{\text{WMMSE}} = \mathcal{O}(NK^2 + I_W K^3). \quad (92)$$

V. ALGORITHM COMPARISON AND SELECTION

The considered analog-beamformer and surface-shaping design is a mixed discrete–continuous and nonconvex problem involving the desired gains, the discrete analog beamformers, and the bounded FIM deformation variables. To exploit the different structures of these variable blocks, we adopt alternating optimization as the overall framework. Specifically, the desired-gain block admits a closed-form solution, the discrete analog-beamformer block is handled by the proximal-distance algorithm, and the bounded deformation block is optimized through a tanh-reparameterized gradient method. Therefore, alternating optimization is not an alternative to the tanh-based method; rather, the latter serves as the solver for the deformation subproblem within the overall AO procedure.

For fixed analog beamformers and desired gains, the deformation subproblem is expressed as

$$\min_{\Delta \mathbf{y}} J(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}) \quad \text{s.t.} \quad -c \leq \Delta y_n \leq c, \quad (93)$$

where $c = \hat{y}/2$. We introduce the reparameterization

$$\Delta y_n = c \cdot \tanh(\alpha z_n), \quad (94)$$

which guarantees deformation feasibility for every finite z_n while preserving the differentiability of the objective. Let

$$\mathbf{D}(\mathbf{z}) = \text{diag}\{c \cdot \alpha \cdot \text{sech}^2(\alpha z_n)\}_{n=1}^N. \quad (95)$$

The transformed gradient satisfies

$$\nabla_{\mathbf{z}} \tilde{J}(\mathbf{z}) = \mathbf{D}(\mathbf{z}) \nabla_{\Delta \mathbf{y}} J(\Delta \mathbf{y}, \mathbf{V}, \mathbf{g}). \quad (96)$$

Consequently, a gradient step in the auxiliary-variable domain induces the following first-order physical-domain update:

$$\Delta \mathbf{y}^{(\ell+1)} - \Delta \mathbf{y}^{(\ell)} = -\eta_\ell \mathbf{D}^2(\mathbf{z}^{(\ell)}) \nabla_{\Delta \mathbf{y}} J + \mathcal{O}(\eta_\ell^2). \quad (97)$$

Thus, the tanh transformation provides not only automatic feasibility but also state-dependent element-wise scaling. In particular, the effective update is smoothly attenuated when a deformation component approaches its physical boundary, thereby avoiding an abrupt clipping operation. Together with the Armijo backtracking line search, this formulation provides a differentiable and computationally efficient way to handle the wavelength-scale deformation constraint. The adopted tanh mapping and backtracking update are described in the surface-shaping block of the proposed algorithm.

Projected gradient descent is also a valid approach because projection onto the box $[-c, c]^N$ can be performed element-wise. Moreover, PGD directly covers the closed feasible interval and permits a convenient evaluation of the projected-gradient stationarity residual. However, direct PGD applies a common scalar step size to all deformation components and employs hard clipping whenever a trial point exceeds the deformation box. For the considered near-field objective, the sensitivities and local curvatures associated with different FIM elements can be highly nonuniform. As a result, a step size that stabilizes the most sensitive elements may lead to slow progress for the remaining elements, whereas a larger step size may cause frequent boundary projections and active-set changes. The tanh-based formulation is therefore selected for its smooth constraint handling and implicit element-wise scaling. Nevertheless, it is not universally superior to PGD: the transformed problem remains non-convex, its solution can depend on initialization, exact boundary points cannot be attained using finite auxiliary variables, and gradient attenuation may occur when the tanh function enters the saturation region. Accordingly, convergence to a first-order stationary solution, rather than global optimality, is claimed.

Manifold optimization is less naturally suited to the constraints considered here. The surface shaping set $[-c, c]^N$ is a box with boundaries rather than a smooth manifold without boundary, while the finite PS set is discrete and therefore does not possess the differentiable geometric structure required by conventional manifold optimization. One possible alternative is to relax the discrete phases to continuous unit-modulus variables and solve the relaxed problem on a product manifold. However, the resulting solution must still be quantized, which may introduce a mismatch between the relaxed optimization problem and the implemented finite-resolution beamformer.

Successive convex approximation is another possible solution strategy. It could locally approximate the nonlinear near-field phase terms and explicitly retain the deformation constraints. However, a suitable convex surrogate must be reconstructed and solved at every iteration. Owing to the coherent coupling among the FIM elements and the exponential dependence of the array response on the deformation variables, constructing a tight, globally valid, and computationally inexpensive surrogate is nontrivial. Moreover, SCA would still generally provide only a local stationary solution. In contrast, the adopted tanh-gradient method requires only objective and gradient evaluations, whose complexity is linear in the numbers of FIM elements and users per gradient iteration.

To complement the optimization-based design, we further propose the key observation method as a non-iterative low-complexity alternative. KOM approximates the coupled surface-shaping problem by independent per-element quartic-polynomial problems. For each FIM element, the globally optimal solution of the simplified scalar problem is obtained by evaluating the objective at all feasible real roots of the corresponding cubic derivative equation and at the two deformation boundaries. Since

the resulting surrogate is separable across the FIM elements, independently combining the scalar minimizers yields the global optimum of the complete separable KOM surrogate.

The global-optimality statement for KOM, however, applies only to the simplified separable surrogate. The original surface-shaping objective contains cross-element coherent terms that couple the deformation variables of different FIM elements. KOM does not explicitly optimize these coupling terms and is therefore not guaranteed to be globally optimal for the original joint problem. In contrast, the AO-based method directly updates the original coupled objective but, owing to its nonconvexity, generally converges only to a block-wise stationary point.

The two proposed methods consequently target different operating requirements. The AO-based tanh method accounts for the coupled multi-beam objective and is intended to provide improved surface-shaping performance at the cost of iterative optimization. KOM avoids AO, PDA, gradient, and line-search iterations and has an overall complexity of $\mathcal{O}(NK)$, making it more suitable for latency- or computation-constrained implementations. Therefore, the proposed methods provide complementary performance–complexity trade-off rather than competing under identical design objectives.

In summary, AO is adopted to decompose the original mixed discrete–continuous problem, while the tanh-based gradient method is selected for the bounded deformation block because of its automatic feasibility, smooth boundary handling, and element-wise adaptive scaling. KOM is further introduced as a non-iterative alternative that sacrifices the explicit treatment of cross-element coupling in exchange for substantially lower computational complexity.